2.5 Multiple-Angle and Product-to-Sum Formulas

Name _____ Period ____ Date _____

Pre-Calculus 2.5 · Student sheet

I can…

- use double-angle formulas to rewrite, evaluate, and solve.
- use power-reducing and half-angle formulas.
- use product-to-sum and sum-to-product formulas.

Vocabulary · fill in the missing word

Double-angle formulas — $\sin 2u =$ _____; $\cos 2u = \cos^2 u - \sin^2 u = 2\cos^2 u - 1 = 1 - 2\sin^2 u$; $\tan 2u = \dfrac{2\tan u}{1 - \tan^2 u}$

Power-reducing formulas — $\sin^2 u =$ _____; $\cos^2 u = \dfrac{1 + \cos 2u}{2}$

Half-angle formulas — $\sin\left(\frac{u}{2}\right) =$ _____; $\cos\left(\frac{u}{2}\right) = \pm\sqrt{\dfrac{1 + \cos u}{2}}$; $\tan\left(\frac{u}{2}\right) = (1 - \cos u)/\sin u$

Product-to-sum — _____ a product like $\sin u \cos v$ as a sum: $\frac{1}{2}[\sin(u + v) + \sin(u - v)]$

Part 1 · Double or not?

Use a calculator in degree mode. Round to three decimal places.

Expression	Value	Compare with	Value	Equal?
$\sin 2(30°)$		$2 \sin 30°$		
$\sin 60°$		$2 \sin 30° \cos 30°$		
$\cos 60°$		$\cos^2 30° - \sin^2 30°$		
$\cos 60°$		$1 - 2 \sin^2 30°$		

So $\sin 2u =$ _____ and $\cos 2u =$ _____

Part 2 · Notes: fill in the blanks

Double-angle: $\sin 2u =$ _____ $\tan 2u =$ _____

$\cos 2u =$ _____ = _____ = _____

Power-reducing: $\sin^2 u =$ _____ $\cos^2 u =$ _____

Half-angle: $\sin(u/2) =$ _____ $\cos(u/2) =$ _____ $\tan(u/2) =$ _____

The ± sign in a half-angle formula depends on the quadrant of _____.

Product-to-sum: $\sin u \cos v =$ _____

Part 3 · Examples (work them with the class)

Example 1

Find all solutions in $[0, 2\pi)$: $2 \cos x + \sin 2x = 0$

Example 2

$\cos u = \frac{5}{13}$ with $\frac{3\pi}{2} < u < 2\pi$. Find sin 2u, cos 2u, and tan 2u.

Example 3

Use a double-angle formula to rewrite 6 sin x cos x.

Example 4

Rewrite $\sin^4 x$ in terms of first powers of cosines of multiple angles.

Example 5

Find the exact value of sin 105°.

Example 6

a. Rewrite cos 5x sin 4x as a sum or difference.

b. Rewrite sin 5x + sin 3x as a product.

Part 4 · Partner practice

Partner 1

$\sin u = \frac{3}{5}$ with $\frac{\pi}{2} < u < \pi$. Find sin 2u.

Partner 2

Find the exact value of $\cos 22.5°$.